

Ryan Matheu | Curriculum Vitae

University of Maryland – College Park
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Education

University of Maryland

Pursuing PhD in Computer Engineering

- Research Advisors: Prof. John S. Baras and Prof. Calin Belta
- Clark Doctoral Fellow

College Park, MD

Aug. 2022 – Present

University of Maryland

BS Physics & BS Mechanical Engineering with Honors

College Park, MD

Graduated Dec. 2021

Industry Experience

Lockheed Martin Advanced Technology Laboratories (ATL)

A/AI Research Engineer Associate

Arlington, VA

Oct. 2025 – Present

University Research Foundation (URF)

Research Engineer Intern

Greenbelt, MD

May 2021 – Aug. 2022

Research Experience

Institute for Systems Research (ISR)

Graduate Research Assistant

College Park, MD

Sept. 2023 – Present

Publications

Peer-reviewed Conference Proceedings

- [C1] R. Matheu, A. G. Puranic, J. S. Baras, and C. Belta, “Bt2automata: Expressing behavior trees as automata for formal control synthesis,” in *Proceedings of the 28th ACM International Conference on Hybrid Systems: Computation and Control*, 2025, pp. 1–11.
- [C2] R. Matheu, A. G. Puranic, J. S. Baras, and C. Belta, “Omtbt: Online monitoring of temporal behavior trees with applications to closed-loop learning,” in *2025 European Control Conference (ECC)*, IEEE, 2025, pp. 2129–2135.

Preprints

- [A1] R. Matheu, J. S. Baras, and C. Belta, “Ternary logic encodings of temporal behavior trees with application to control synthesis,” *arXiv preprint arXiv:2604.12092*, 2026, (Accepted CDC 2026).
- [A2] R. Matheu, S. S. Damera, J. S. Baras, and C. Belta, “Differentiable ternary temporal logic semantics with polynomial surrogate networks,” *arXiv preprint arXiv:2610.04662*, 2026.
- [A3] S. S. Damera, R. Matheu, A. G. Puranic, and J. S. Baras, “Polynomial surrogate training for differentiable ternary logic gate networks,” *arXiv preprint arXiv:2603.00302*, 2026, To appear in Proceedings of 3rd International Conference on Neuro-Symbolic Systems (NeuS).
- [A4] S. S. Damera, R. Matheu, A. G. Puranic, J. S. Baras, and C. Belta, “On the stability and realizability of recurrent polynomial surrogate ternary logic gate networks,” *arXiv preprint arXiv:2605.24649*, 2026, (Accepted CDC 2026).
- [A5] J. S. Baras, S. S. Damera, R. Matheu, C. Enwerem, and P. Kumar, “Iddmbse: Integrating data-driven and model-based systems engineering for trusted autonomous cyber-physical systems,” *arXiv preprint arXiv:2606.06727*, 2026, (Accepted IEEE ISSE 2026).

Talks

Contract-Based Agentic Systems Engineering: Delegation as Quotient for Trusted Cyber-Physical System Design

7th Annual AI4SE & SE4AI Research and Application Workshop

Sept. 2026

Teaching Experience

Guest Lectures

ENEE 467: Robotics Project Laboratory

– Presented three lectures on programming robots with ROS2.

College Park, MD

Sept. 2025

Teaching Assistant

ENEE 660: Linear Systems Theory

ENEE 245: Digital Circuits and Systems

ENEE 303: Analog and Digital Electronics

ENEE 476: Renewable Energy

College Park, MD

Aug. – Dec. 2023

Aug. – Dec. 2023

Jan. – May. 2023

Aug. – Dec. 2022

Qualifications, Achievements, & Awards

Clark Doctoral Fellows Mid-Career Award

Apr. 2026

NSF Student Travel Award

CPS-IoT Week 2025

EIT Certified, Mechanical

State of Maryland

– License No. 59754

Aug. 2022

Academic and Professional Service

Session Co-Chair, *Safety Critical Systems*, European Control Conference (ECC) *2025*

Refereed papers (reviewer) for the following journals and conferences:

– IEEE Transactions on Automatic Control (IEEE Trans. Autom. Control) *2026*

– Subreviewer/Delegated Reviewer:

★ IROS *2026*

Technical Skills

Robotics: ROS2, Mujoco, Isaac Sim

Optimization: Gurobi, CVXPY, Pyomo

Model Checking: UPPAAL, Prism, nuXmv

Provers/Solvers: Lean, Z3

GPU Computing: JAX, Warp

APIs: Agents SDK, LangGraph